Response to the Reviewer Comments on Manuscript "Federated Wind Speed Forecasting in Finland: Geographic vs. Correlation Graphs with Synchronous FedGD"

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We thank the reviewers and the teaching assistant for their constructive and detailed feedback. We have addressed all raised issues in the revised manuscript.

Major modifications include:

- The report title was modified to describe the technical approach of the project better. Previously it only mentioned the application domain, but I added the main idea (comparing graph structures) and the methods (FL with synchronous FedGD) to the title.
- All plot numbering from the figures removed and the latex auto-generated plot numbering is now used for referencing figures and plots.
- The feature count was corrected from 7 to 6, more mathematical notation was added where appropriate, and more justification for the use of the ℓ_2 norm was added.
- The future work section was expanded from a short paragraph into a list of bullet points.

In the following, we respond to each comment point by point. Section, figure, and equation numbers in the quoted comments refer to the original manuscript; numbers in our responses refer to the revised manuscript.

Comments of Reviewer #1

1.1 *The report should include the list of FMI station IDs or names used (e.g., as a table or supplementary reference), since the ≥80% completeness filter applied to an undated API call cannot reliably reproduce the exact 50 stations.*

- A complete station list was omitted given the 136 stations involved. Instead, the data query description in the numerical experiments section was strengthened with the exact API query, bounding box, variable names, and processing pipeline, so the filtering procedure can be reproduced exactly.

1.2 *There is an inconsistency where figure numbers embedded inside plots (e.g., "Figure 4", "Figure 7", "Figure 8") do not match the caption labels below them ("Fig. 2", "Fig. 5", "Fig. 6"): these should be cleaned up before the final submission.*

- All in-plot titles were removed. Now all figures only use the latex autogenerated figure numbering.

1.3 *Some of the texts are overlapping and affect the readability of plots (Fig. 6 / Figure 8). Since there are already captions below each figure, the captions inside the plots seem unnecessary and reduce legibility.*

- The plot text and titles were simplified so that the caption below each figure conveys the necessary information.

Comments of Reviewer #2

2.1 *The report discusses the use of 7 features, but it only explains 6: 3 lagged wind speeds (t-1, t-2, t-3), 1 lagged atmospheric pressure, 1 lagged temperature, 1 lagged humidity.*

- The feature count was corrected to 6 throughout the manuscript. The weight vector $\mathbf{w}^{(i)} \in \mathbb{R}^7$ has 7 components because the feature vector is augmented with a bias term, not because there are 7 features.

2.2 *Feature normalization is repeated unnecessarily in the Problem Formulation section.*

- The repeated normalization description was removed from the Problem Formulation section. Feature and label normalization are now described only once, in Section 2.

2.3 *GTVMin (and FedGD) is discussed in Section 2 before being explained in Section 3. It is also not cited. The reader should not be assumed to be previously familiar with GTVMin.*

- A forward reference "(defined in Section 3)" and a citation were added at the first mention of GTVMin in Section 2. FedGD is also now cited when mentioned the first time.

2.4 *For the Geographic k-NN system, the edge weight kernel explanation is a bit unclear: it uses a Gaussian kernel (not mentioned or cited); the report makes it sound like all of Equation 2 is the Haversine distance; the Haversine formula is not shown or cited.*

- The edge weight formula (Eq. 2) is now described correctly as a Gaussian kernel.
- The variable d_{ij} in Eq. 2 is now clearly explained as the Haversine distance between stations i and j, so it is clear that the Haversine formula gives d_{ij} and not the full equation.
- A citation to the Haversine formula was added.

2.5 *Some subsections would improve the report's readability.*

- Subsections were not added. The spconf IEEE template is a well recognized template and custom subsections might actually hurt readability if readers are expecting a certain template structure.

2.6 *The figures have inconsistent number labeling in their titles and in the report.*

- All in-plot figure numbering removed.

Comments of Teaching Assistant (Alexander Pavlyuk)

TA.1 *Criterion 1: The title is not informative. Please indicate (i) the main idea (comparing graph structures), (ii) the application domain (wind prediction), and (iii) the methods (FL, Synchronous FedGD).*

- The title was changed to "Federated Wind Speed Forecasting in Finland: Geographic vs. Correlation Graphs with Synchronous FedGD". It now conveys (i) the comparison of graph structures (Geographic vs. Correlation Graphs), (ii) the application domain (wind speed forecasting in Finland), and (iii) the method (Synchronous FedGD).

TA.2 *Criterion 6: The label was not explicitly defined. Mathematical notation should be added to improve readability.*

- The prediction label is now formally defined as $y_t^{(i)} = ws_{t+1}^{(i)}$ (next-hour wind speed at station i) at the start of Section 2.
- The full feature vector is now written out as a math expression with all six components named and their physical units stated.

TA.3 *Criterion 9: More arguments in favor of the squared ℓ_2 norm can be made.*

- Two more arguments for the squared ℓ_2 norm were added in Section 3:
  a) it penalizes large parameter disagreements quadratically, encouraging more agreement between neighbors
  b) adding a strictly convex penalty to the sum of convex local losses makes the full GTVMin objective strongly convex for $\alpha > 0$, guaranteeing a unique global minimizer

TA.4 *Criterion 14: Figure title/numbering issues; y-axis label out of bounds on Figure 3; in-plot title overlaps; station coordinates used instead of names.*

- All in-plot figure numbers were removed and now the report only relies on latex automatic figure numbering
- The y-axis label that was clipped out of bounds on Fig. 3 was moved to be in frame.
- Station coordinates used to label the best and worst performing stations were replaced with their FMI station names for readability.

TA.5 *Criterion 16: I strongly suggest including more detailed potential improvements and presenting them in a list.*

- The future directions were rewritten from a short paragraph into a bullet list.
- The future directions contain reasons why the directions would strengthen the understanding of the application.